Introduction to Machine Learning

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1 Introduction to Probability Theory

In engineering applications, uncertainty arises from incomplete knowledge, measurement errors, variability in material properties and loads, and the intrinsic complexity of physical systems. This chapter introduces the fundamental concepts of probability theory, which provides the mathematical framework for describing and quantifying uncertainties and for reasoning about their consequences.

1.1 Discrete and Continuous Random Variables

Let us start with the mathematical description of an uncertain experiment. An experiment is a process whose outcome can not be predicted with certainty before it is performed. The possible outcomes are collected in a *sample space* Ω. An *event* is a collection of possible outcomes, i.e., a subset of the sample space.

For example, consider the experiment of measuring the compressive strength of a concrete specimen, which is described by a sampling space $\Omega = \{\texttt{low, ordinary, high}\}$. The outcome of the experiment is the measured strength, which may take different values depending on the specimen itself and the experimental conditions. A possible event could be defined as the compressive strength exceeding a prescribed threshold $A_1 = \{\texttt{high}\}$. Another possible event could be defined as the compressive strength *not* exceeding a prescribed threshold $A_2 = A_1^c = \{\texttt{low, ordinary}\}$

Probability of an Event. A probability measure P assigns a number between 0 and 1 to each event, quantifying the degree of uncertainty/certainty associated with its occurrence. The probability measure satisfies the following three axioms:

- Axiom I (non-negativity). For any event A, the probability $P(A)$ is a real number greater than or equal to zero:
$$P(A) \geq 0. \tag{1.1}$$

- Axiom II (normalization). The probability of the entire sample space Ω (the set of all

possible outcomes, i.e., certain event) is unitary:

$$P(\Omega) = 1. \tag{1.2}$$

- Axiom III (countable additivity). For any collection of events $A_1, A_2, \ldots, A_N$ that are mutually exclusive, i.e., $A_n \cap A_m = \emptyset$, $\forall n \neq m$ with $n, m = 1, \ldots, N$, the probability of their union is the sum of their individual probabilities:

$$P\left(\bigcup_{n=1}^{N} A_n\right) = \sum_{n=1}^{N} P(A_n). \tag{1.3}$$

Some useful properties can be derived from these axioms:

- The probability of an impossible event (empty set) vanishes

$$P(\emptyset) = 0; \tag{1.4}$$

- If event A is a subset of event B ($A \subseteq B$), then monotonicity holds

$$P(A) \leq P(B); \tag{1.5}$$

- The probability of an event not happening is given by the complement rule

$$P(A^c) = 1 - P(A), \tag{1.6}$$

where A^c is the complement of an event A;

- The probability of any event A is always bounded as

$$0 \leq P(A) \leq 1; \tag{1.7}$$

- For two arbitrary events A and B, not necessarily disjoint, the probability of their union is given by the addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \tag{1.8}$$

where subtracting the last term prevents the intersection of the two events from being counted twice, also known as inclusion-exclusion principle.

Random Variables. Events provide a way to describe uncertain outcomes qualitatively. To represent these uncertainties numerically, we introduce the *random variable* construct $X : \Omega \mapsto \mathbb{R}$, which assigns a numerical value to each outcome in the sample space. The capitalized notation X is conventionally used for the random variable, representing the uncertain quantity that can take different values. The lowercase x denotes one of its possible realizations, once the experiment is performed and a specific value x is observed. For example, the compressive strength of a *random* concrete specimen can be represented by a random variable X, while the *measured strength of a particular specimen* is a realization x of X.

The sampling space for a random variable can either be discrete or continuous. A random variable is *discrete* if it can assume a finite or countably infinite number of distinct values. Conversely, a random variable is *continuous* if it can take values over a continuous interval of the real line.

Discrete Random Variables. For a discrete random variable, uncertainty is described by a *probability mass function* (PMF), defined as

$$p_X(x) := P(X = x), \tag{1.9}$$

yielding the probability that the random variable assumes the specific value $X = x$. By the probability axioms, since the possible values of X are mutually exclusive and exhaustive, the probabilities satisfy the following relations:

$$p_X(x) \geq 0, \qquad \sum_x p_X(x) = 1. \tag{1.10}$$

Similarly, the probability that X belongs to a set (event) A is obtained by summing the probabilities of all possible realizations contained in that set:

$$P(X \in A) = \sum_{x \in A} p_X(x). \tag{1.11}$$

Continuous Random Variables. For a continuous random variable the distribution is instead described by a *probability density function* (PDF) $p_X(x)$. Note that the PDF $p_X(x)$ is not itself a probability, as any individual value x has zero probability for a continuous random variable, i.e., $P(X = x) = 0$. Probabilities are instead obtained by integrating the PDF over finite intervals. In particular, the probability of observing a value within an interval $[a, b]$ is

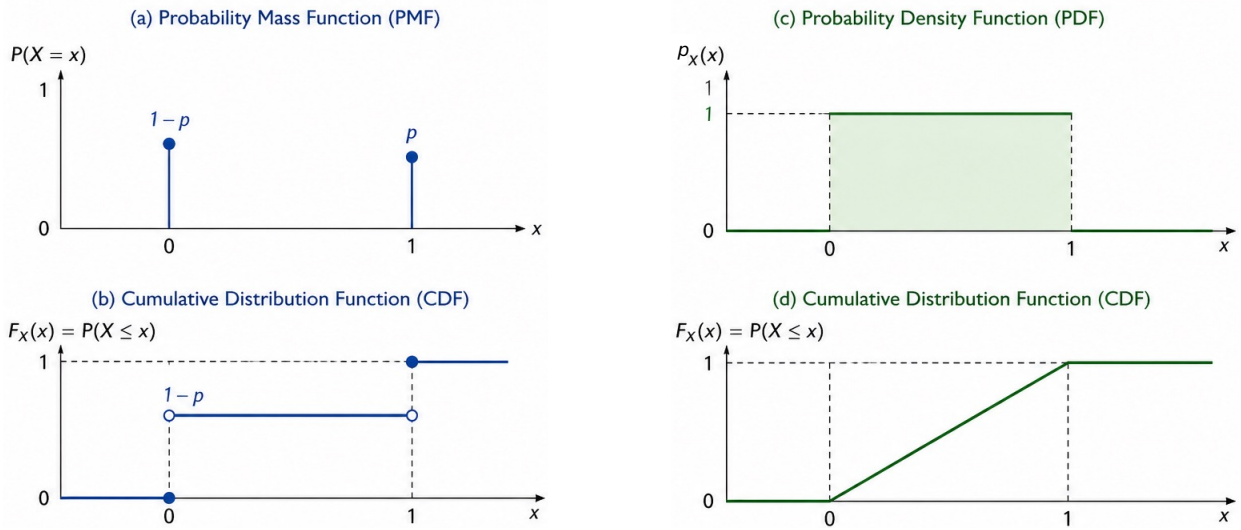


Figure 1.1: Comparison of discrete and continuous random variables.

given by:

$$P(a \leq X \leq b) = \int_a^b p_X(x)\,\mathrm{d}x. \tag{1.12}$$

Following the probability axioms, the PDF satisfies

$$p_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} p_X(x)\,\mathrm{d}x = 1. \tag{1.13}$$

Cumulative Distribution Function. Both discrete and continuous random variables admit a *cumulative distribution function* (CDF), defined as

$$F_X(x) = P(X \leq x), \tag{1.14}$$

which provides the probability that the random variable assumes a value smaller than or equal to a given x.

For a discrete random variable, the CDF takes the name of *cumulative mass function* and

is obtained by summing the probabilities that the PMF assigns to individual values:

$$F_X(x) = \sum_{\xi \leq x} p_X(\xi). \tag{1.15}$$

Inversely, the PMF can be obtained from the CDF, as follows:

$$p_X(x_i) = F_X(x_i) - F_X(x_{i-1}). \tag{1.16}$$

For a continuous random variable, the CDF takes the name of *cumulative density function* and is obtained by integrating the density of probability over the real line, as specified by the PDF, as follows:

$$F_X(x) = \int_{-\infty}^{x} p_X(\xi)\, \mathrm{d}\xi. \tag{1.17}$$

Inversely, whenever the derivative exists, the PDF is obtained by differentiating the CDF:

$$p_X(x) = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x}. \tag{1.18}$$

It is worth noting that since $F_X(x)$ is the integral of $p_X(x)$ and $p_X(x) \geq 0$, it follows that $F_X(x)$ increases monotonically, with values at the lower and upper bounds $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$, respectively.

1.2 Conditional Probability and Independence

Conditional probability provides the formalism to describe the relationships between random variables, without requiring any causal interpretation. Conditioning on an event or a random variable simply means restricting the probabilistic description to the information that is currently available. As we will see, this concept is fundamental to the Bayes' theorem, as it enables the mechanism for updating the probability assigned to an event when additional information becomes available.

Conditional Probability. Consider two events A and B defined on the same sample space Ω. If $P(B) > 0$, the probability of A conditional on the occurrence of B (conditional probability of A given B) is defined as

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \tag{1.19}$$

It is worth noting how this notion of probability does not explicitly assume nor enforce any causal effect or time-ordering, e.g., it does not by itself mean that B causes A. Rather, it is the probability assigned to A after restricting the subset of the sample space in which B occurs. In general, this changes the probability assigned to an event, e.g., $P(A \mid B) \neq P(A)$.

Equivalently, the definition can be rearranged as

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A), \tag{1.20}$$

also known as *product rule* or *chain rule of probability*, according to which the joint probability of N events can be broken down into $N - 1$ conditionals and one marginal probability.

Conditional probability can be extended from events to random variables, where it provides a probabilistic description of one variable when the value of another variable is known. For example, given two continuous random variables X and Y, respectively representing the distribution of an uncertain material parameter and of a structural response, knowledge of $X = x$ then modify the probabilistic prediction of Y, as expressed through the conditional PDF of Y given X as follows:

$$p_{Y|x}(y \mid x) = \frac{p_{X,Y}(x, y)}{p_X(x)}, \qquad p_X(x) > 0, \tag{1.21}$$

where $p_{X,Y}(x, y)$ denotes the joint PDF of X and Y and $p_X(x)$ is the marginal PDF of X. The joint probability density can thus be factorized as

$$p_{X,Y}(x, y) = p_{Y|x}(y \mid x)p_X(x). \tag{1.22}$$

An equivalent result holds for the conditional PMF of two discrete random variables.

Law of Total Probability. Conditional probabilities can be used to recover an unconditional probability by averaging over all possible conditioning events or values. Suppose that $\{A_1, \ldots, A_N\}$ forms a partition of the sample space, meaning that the events are mutually exclusive ($A_n \cap A_m = \emptyset$, $\forall n \neq m$ with $n, m = 1, \ldots, N$) and exhaustive ($\bigcup_{n=1}^{N} A_n = \Omega$). Then the probability of any event B can be obtained as follows:

$$P(B) = \sum_{n=1}^{N} \underbrace{P(B \mid A_n)P(A_n)}_{P(B \cap A_n)}, \tag{1.23}$$

which is known as the *law of total probability*.

The continuous counterpart is obtained by conditioning on the values of a continuous random variable X:

$$P(B) = \int_{-\infty}^{\infty} P(B \mid X = x) p_X(x)\, \mathrm{d}x. \tag{1.24}$$

Similarly, for a continuous random variable Y, the marginal PDF can be obtained as follows:

$$p_Y(y) = \int_{-\infty}^{\infty} p_{Y|x}(y \mid x) p_X(x)\, \mathrm{d}x. \tag{1.25}$$

showing that the marginal distribution of Y can be interpreted as an average of the conditional distributions $p_{Y|x}(y \mid x)$ weighted according to the probability distribution of X.

Independence. Two events A and B are said to be *independent* if the occurrence of one event does not alter the probability of the other, i.e., $P(A \mid B) = P(A)$, provided that $P(B) > 0$. Combining this relation with the product rule gives the following equivalent condition for the joint probability of independent events:

$$P(A \cap B) = P(A \mid B)P(B) = P(A)P(B). \tag{1.26}$$

The concept similarly applies to random variables. Two (discrete or continuous) random variables X and Y are independent if the conditional distribution of one variable does not depend on the value of the other, i.e., $p_{Y|x}(y \mid x) = p_Y(y)$. Their joint distribution thus factorizes into the product of their marginal distributions:

$$p_{X,Y}(x, y) = p_{Y|x}(y \mid x) p_X(x) = p_Y(y) p_X(x). \tag{1.27}$$

For independent random variables, knowledge of X provides no information about Y. However, when X and Y are not independent, their joint distribution can not be expressed as the product of their marginal distributions, and conditioning on one variable changes the distribution of the other.

Remark. It is important to distinguish *independence* from *uncorrelation*. Independence implies zero covariance $\mathrm{Cov}(X, Y) = 0$ but the converse is not true in general. Two random variables may be uncorrelated while exhibiting statistical dependence. An important exception is represented by jointly Gaussian random variables, for which zero covariance is sufficient to establish independence.

A related important concept is *conditional independence*. Two random variables X and Y are conditionally independent given a third random variable Z if

$$p_{X,Y|z}(x, y \mid z) = p_{X|z}(x \mid z)p_{Y|z}(y \mid z). \tag{1.28}$$

An equivalent statement is as follows:

$$p_{X|y,z}(x \mid y, z) = p_{X|z}(x \mid z), \tag{1.29}$$

showing that, once Z is known, additional knowledge of Y does not provide information about X. This notion is largely employed for representing probabilistic models, e.g., in probabilistic graphical models, as it allows a high-dimensional joint distribution to be decomposed into simpler conditional distributions.

1.2.1 Example: Post-Earthquake Structural Assessment

Consider the following example adapted from [1]. Let the health state of a structure after an earthquake be represented by a discrete random variable X, and let the peak ground acceleration (PGA) during the earthquake be represented by a continuous random variable Y. The conditional probability of a compromised structural state given the observed PGA can then be used to *infer* the presence of structural damage without directly observing the state of the structure.

By rewriting the conditional probability in Eq. (1.21), the (posterior) probability of a structural state x given an observed PGA value y is obtained as follows:

$$p_{X|y}(x \mid y) = \frac{p_{Y|x}(y \mid x)p_X(x)}{p_Y(y)}. \tag{1.30}$$

Herein, $p_Y(y)$ is the marginal probability density of the observed PGA and acts as a normalization constant ensuring that the posterior probabilities sum to one. Since X is discrete, the marginal density of Y is obtained by summing the joint density over all possible structural states:

$$p_Y(y) = \sum_{x \in \Omega} f_{Y|x}(y \mid x)p_X(x), \tag{1.31}$$

which follows from the law of total probability in Eq. (1.23), and where Ω denotes the set of possible structural states.

Substituting this expression into Eq. (1.30) we obtain:

$$p_{X|y}(x \mid y) = \frac{p_{Y|x}(y \mid x)p_X(x)}{\sum_{x' \in \Omega} p_{Y|x}(y \mid x')p_X(x')}. \tag{1.32}$$

The posterior probability of each structural state is thus obtained by multiplying its prior probability $p_X(x)$ by the conditional probability density $p_{Y|x}(y \mid x)$ of observing the measured PGA value y, given that structural state x, and then normalizing by the probability $p_Y(y)$ of the observation itself.

The prior probability $p_X(x)$ represents the knowledge about the structural state before considering the PGA observation, while the conditional probability density $p_{Y|x}(y \mid x)$ describes how compatible the observed earthquake intensity is with each possible structural state. The PGA observation therefore modifies the initial prior, producing a posterior probability distribution over the structural states.

This kind of evaluations are useful when the actual structural state can not be directly observed. Indeed, the measured PGA can provide information about the probability of damage and can consequently be used to support structural safety assessment and decision-making.

1.3 Multivariate Random Variables

In many situations, we are required to deal with multiple uncertain quantities simultaneously. For example, the uncertain compressive strength of concrete depends on the water content, cement content, age, and other material properties. In such cases, describing each quantity independently is generally insufficient because the variables may exhibit statistical dependence, reflecting coupling effects. Multivariate modeling enables describing the joint behavior of multiple random variables and their relationships.

Random Vectors and Joint Probability Distributions. Given d random variables, these can be collected into a *random vector* $\mathbf{X} = [X_1, X_2, \ldots, X_d]^\top \in \mathbb{R}^d$. A realization of the random vector is simply denoted by $\mathbf{x} = [x_1, x_2, \ldots, x_d]^\top$, collecting the realizations of each corresponding random variable. The probabilistic behavior of $\mathbf{X}$ is described by its *joint probability distribution*, which characterizes the probability of observing different combinations of values of the random variables.

As an example for $d = 2$, given two discrete random variables X and Y, the joint PMF

is defined as

$$p_{X,Y}(x, y) := P(X = x, Y = y), \tag{1.33}$$

expressing the probability that $X = x$ while simultaneously $Y = y$.

The counterpart for continuous random variables is given by the joint PDF $p_{X,Y}(x, y)$. The probability that (X, Y) belongs to a certain region $\mathcal{A} \subseteq \mathbb{R}^2$ is obtained by integrating the joint density, as follows:

$$P\left((X, Y) \in \mathcal{A}\right) = \int_{\mathcal{A}} p_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y, \tag{1.34}$$

where, following the probability axioms, the joint PDF satisfies

$$p_{X,Y}(x, y) \geq 0, \qquad \int_{\mathbb{R}^2} p_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y = 1. \tag{1.35}$$

For a generic random vector $\mathbf{X}$, the multivariate (joint) CDF is defined as follows:

$$F_{\mathbf{X}}(\mathbf{x}) = P(X_1 \leq x_1, \ldots, X_d \leq x_d). \tag{1.36}$$

As in the univariate case, the CDF completely characterizes the probability distribution. For continuous random variables, the joint PDF can be obtained by differentiating the joint CDF:

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^d F_{\mathbf{X}}(\mathbf{x})}{\partial x_1 \partial x_2 \ldots \partial x_d}. \tag{1.37}$$

Marginal and Conditional Distributions. While the joint distribution characterizes the probabilistic behavior of the random vector, the distribution of an individual variable can be recovered by *marginalizing* with respect to the remaining variables. For example, given the joint PMF $p_{X,Y}(x, y)$ of two discrete random variables X and Y, the marginal PMF of X is obtained by summing over all possible values of Y, as follows:

$$p_X(x) = \sum_y p_{X,Y}(x, y). \tag{1.38}$$

For example, consider two binary random variables X and Y, whose joint PMF is given

by the following table:

$$p_{X,Y}(x,y) = \left\{ \begin{array}{c|cc|c} & Y=0 & Y=1 & p_X(x) \\ \hline X=0 & 0.30 & 0.20 & 0.50 \\ X=1 & 0.10 & 0.40 & 0.50 \\ \hline p_Y(y) & 0.40 & 0.60 & 1.00 \end{array} \right.$$

For instance, the marginal probability of $X = 0$ is obtained by summing the joint probabilities over the two possible values of Y:

$$p_X(0) = p_{X,Y}(0,0) + p_{X,Y}(0,1) = 0.30 + 0.20 = 0.50,$$

and similarly:

$$p_X(1) = p_{X,Y}(1,0) + p_{X,Y}(1,1) = 0.10 + 0.40 = 0.50.$$

Thus, the marginal PMF of X is

$$p_X(x) = \begin{cases} 0.50, & x = 0, \\ 0.50, & x = 1. \end{cases}$$

Analogously, the marginal PMF of Y is obtained by summing over the possible values of X, yielding:

$$p_Y(0) = 0.30 + 0.10 = 0.40, \tag{1.39}$$

$$p_Y(1) = 0.20 + 0.40 = 0.60. \tag{1.40}$$

For continuous random variables with joint PDF $p_{X,Y}(x,y)$, the marginal PDF of X is obtained by integrating out Y, as follows:

$$p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)\,\mathrm{d}y. \tag{1.41}$$

Both operations can be interpreted as "averaging out" the uncertainty associated with the variables that are summed or integrated out. Similarly, for a d-dimensional random vector, the marginal distribution of any subset of variables is obtained by summing or integrating the joint distribution over the remaining variables.

We have already seen as the joint distribution also provides a direct way of obtaining

conditional distributions. For two random variables, the conditional distribution is given by Eq. (1.21). Combining the definitions of the marginal and conditional distributions also gives the factorization in Eq. (1.22).

As introduced in the previous section, for two random variables, independence in terms of the joint distribution is expressed as $p_{X,Y}(x, y) = p_X(x)p_Y(y)$, i.e., knowledge of one does not provide information about the other $p_{Y|x}(y \mid x) = p_Y(y)$. However, in the general case $p_{X,Y}(x, y) \neq p_X(x)p_Y(y)$, and the variables exhibit statistical dependence.

Random Samples and Data Matrices. The random vector $\mathbf{X}$ can represent a collection of measurable features or uncertain parameters, while multiple realizations of this vector can be obtained from experiments or simulations, depending on the cases. Given N observations $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N$, these can be arranged into a data matrix:

$$\mathcal{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times d}, \tag{1.42}$$

where each row corresponds to one realization of the random vector, while each column corresponds to one random variable, also known as covariate or feature depending on the context. This data structure provides a useful representation for many machine learning methods presented in the following chapters.

1.4 Moments and Expectations

Probability distributions fully characterize the uncertainty associated with a random variable. However, it is often useful to summarize a distribution through a small number of quantities that characterize its central tendency, dispersion, and shape. These quantities are known as *moments* and are constructed through the *expectation* operator.

Expectation. The expectation, or expected value, of a random variable can be interpreted as its probability-weighted average. For a discrete random variable X with PMF $p_X(x)$, the expectation is defined as follows:

$$\mathbb{E}[X] = \sum_x x\, p_X(x). \tag{1.43}$$

For a continuous random variable, the corresponding definition is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, p_X(x)\, \mathrm{d}x. \tag{1.44}$$

The expectation operator is linear; given two random variables X and Y and constants a and b, the following relations hold:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \tag{1.45}$$

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]. \tag{1.46}$$

The notion can be extended to any function $g(X)$ of a random variable X, as follows:

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x), \qquad \text{(for discrete variables)} \tag{1.47}$$

$$= \int_{-\infty}^{\infty} g(x) p_X(x)\, \mathrm{d}x. \qquad \text{(for continuous variables)} \tag{1.48}$$

Conditional Expectation. The concept of expectation also extends naturally to conditioned random variables. The conditional expectation is itself a random variable because it depends on other random variables. For a discrete random variable Y, the conditional expectation given $X = x$ reads:

$$\mathbb{E}[Y \mid X = x] = \sum_y y\, p_{Y|x}(y \mid x). \tag{1.49}$$

For a continuous random variable it is given by:

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y\, p_{Y|x}(y \mid x)\, \mathrm{d}y. \tag{1.50}$$

Mean and Variance. The expected value is commonly referred to as the *mean* of the random variable and is denoted by

$$\mu_X = \mathbb{E}[X], \tag{1.51}$$

which measures the central tendency of a probability distribution. It coincides with the centroid of the probability density or mass function, a concept analogous to the center of gravity for a solid body or cross section.

The dispersion of a random variable around its mean is instead quantified by the *variance*, which is defined as follows:

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mu_X)^2\right]. \tag{1.52}$$

The concept of variance is analogous to the notion of moment of inertia for a cross section.

Using the linearity of the expectation operator, the variance can equivalently be written as follows:

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}([X-\mu_X)^2] && (1.53)\\
&= \mathbb{E}[X^2 - 2X\cdot\mathbb{E}[X] + \mathbb{E}[X]^2] && (1.54)\\
&= \mathbb{E}[X^2] - \mathbb{E}[2X\cdot\mathbb{E}[X]] + \mathbb{E}[\mathbb{E}[X]^2] && (1.55)\\
&= \mathbb{E}[X^2] - 2\mathbb{E}[X]\cdot\mathbb{E}[X] + \mathbb{E}[X]^2 && (1.56)\\
&= \mathbb{E}[X^2] - \mathbb{E}[X]^2. && (1.57)
\end{aligned}$$

The square root of the variance $\sigma_X = \sqrt{\mathrm{Var}(X)}$ is called *standard deviation* and it features the same physical units as the random variable itself.

For a linear function $Y = g(X) = aX + b$ of a random variable X, with parameters $a, b \in \mathbb{R}$, because of the linearity of the expectation operation, we have that:

$$\mu_Y = \mathbb{E}[Y] = \mathbb{E}[aX+b] = a\mu_X + b \equiv g(\mu_X), \qquad \sigma_Y^2 = \mathrm{Var}(Y) = a^2\sigma_X. \tag{1.58}$$

Higher-Order Moments. More generally, the rth *central* moment is defined with respect to the mean as $\mathbb{E}\left[(X-\mu_X)^r\right]$.

The first central moment is always zero $\mathbb{E}[X-\mu_X] = 0$, while the second central moment corresponds to the variance. Higher-order central moments provide information about the shape of a probability distribution. In particular, the third central moment is the *skewness*, which is defined as follows:

$$\gamma_{X,3} = \frac{\mathbb{E}[(X-\mu_X)^3]}{\sigma_X^3}, \tag{1.59}$$

characterizing the asymmetry of a distribution. The fourth central moment is instead the *kurtosis*, which is defined as follows:

$$\gamma_{X,4} = \frac{\mathbb{E}[(X-\mu_X)^4]}{\sigma_X^4}, \tag{1.60}$$

describing the relative weight of its tails.

Covariance. For two random variables X and Y, the variance can be generalized to quantify their joint variability through the *covariance*:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)], \tag{1.61}$$

which can equivalently be written as

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]E[Y]. \tag{1.62}$$

It measures the tendency of two random variables to vary together. A positive covariance indicates that large values of one variable tend to occur together with large values of the other, whereas a negative covariance indicates an inverse tendency. A covariance close to zero indicates the absence of a linear relationship, although it does not generally imply statistical independence. On the other hand, if X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, and therefore $\mathrm{Cov}(X, Y) = 0$. However, the converse is generally not true, as zero covariance does not imply statistical independence. Finally, the covariance between a random variable and itself reduces to the variance:

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X). \tag{1.63}$$

Correlation. Since the magnitude of covariance depends on the units and scale of the random variables, it is often useful to introduce the dimensionless *correlation coefficient* capturing the tendency of random variables to vary together, i.e., the linear dependence between two random variables:

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}, \qquad -1 \leq \rho_{X,Y} \leq 1. \tag{1.64}$$

Values close to 1 indicate a strong positive linear relationship, values close to -1 indicate a strong negative linear relationship, and values close to zero indicate weak linear correlation (see the top panel of Fig. 1.2. A correlation coefficient $\rho_{X,Y} \pm 1$ indicates an exact linear relationship between the two variables. On the other hand, $\rho_{X,Y} = 0$ indicates that the variables are uncorrelated. This happens, for instance, when two random variables are independent. However, the inverse is not true, and a correlation coefficient $\rho_{X,Y} = 0$ does not automatically implies independence between X and Y. Indeed, the two random variables can display a nonlinear dependence also when $\rho_{X,Y} = 0$ (see the bottom panel of top Fig. 1.2).

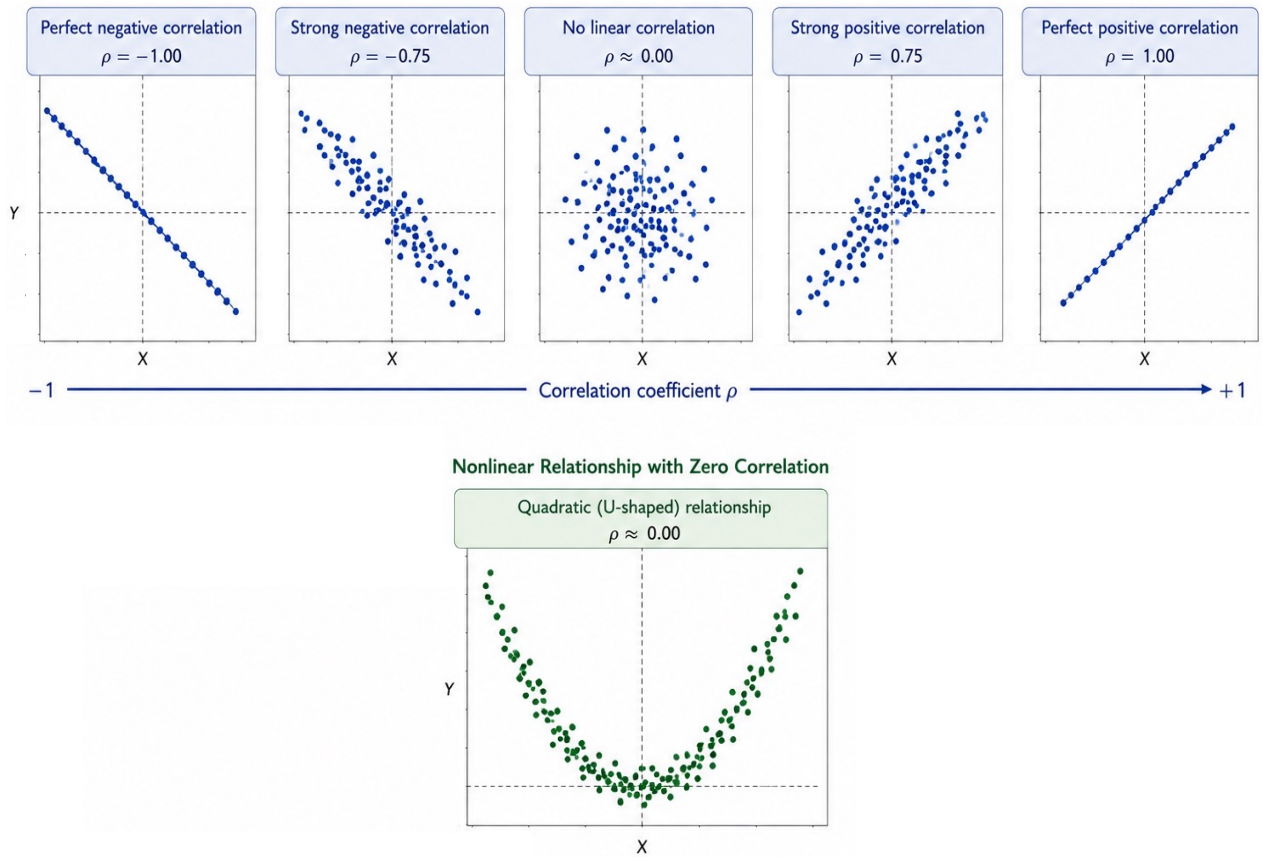


Figure 1.2: Realizations of two random variables for different correlation coefficients.

Moments of a Random Vector. The concept of moments can be similarly extended to random variables. Specifically, the expected value of a random vector $\mathbf{X} \in \mathbb{R}^d$ is defined component-wise, as follows:

$$\mathbb{E}[\mathbf{X}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{bmatrix} = \boldsymbol{\mu}, \tag{1.65}$$

which often takes the name of *mean vector*.

The *covariance matrix* of $\mathbf{X}$ is instead defined as follows:

$$\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top] = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1,X_2) & \cdots & \mathrm{Cov}(X_1,X_d) \\ \mathrm{Cov}(X_2,X_1) & \mathrm{Var}(X_2) & \cdots & \mathrm{Cov}(X_2,X_d) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_d,X_1) & \mathrm{Cov}(X_d,X_2) & \cdots & \mathrm{Var}(X_d) \end{bmatrix} \tag{1.66}$$

$$= \begin{bmatrix} \sigma^2_{X_1} & \rho_{X_1,X_2}\sigma_{X_1}\sigma_{X_2} & \cdots & \rho_{X_1,X_d}\sigma_{X_1}\sigma_{X_d} \\ \rho_{X_1,X_2}\sigma_{X_1}\sigma_{X_2} & \sigma^2_{X_2} & \cdots & \rho_{X_2,X_d}\sigma_{X_2}\sigma_{X_d} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{X_1,X_d}\sigma_{X_1}\sigma_{X_d} & \rho_{X_2,X_d}\sigma_{X_2}\sigma_{X_d} & \cdots & \sigma^2_{X_d} \end{bmatrix}$$

The diagonal entries quantify the variance of each individual random variable, while the off-diagonal entries quantify the pairwise covariance between different variables. By construction, the covariance matrix is symmetric $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top$ and positive semi-definite, i.e., $\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a} \geq 0$, for any $\mathbf{a} \in \mathbb{R}^d$.

Expectation of a Function of a Random Vector. For a response $Y \in \mathbb{R}$ that depends on a continuous random vector $\mathbf{X} \in \mathbb{R}^d$ of uncertain parameters $\mathbf{X}$ through $Y = g(\mathbf{X})$, $g : \mathbb{R}^d \mapsto \mathbb{R}$, the expectation is obtained from the joint probability density, as follows:

$$\mathbb{E}[Y] = \mathbb{E}[g(\mathbf{X})] = \int_{\mathbb{R}^d} g(\mathbf{x}) p_{\mathbf{X}}(\mathbf{x})\, \mathrm{d}\mathbf{x}. \tag{1.67}$$

When this integral cannot be evaluated analytically, it can be approximated numerically, for example using Monte Carlo sampling. This situation reflects many engineering applications where quantities of interest are nonlinear functions of several uncertain variables.

In the case of a linear function $Y = g(\mathbf{X}) = \mathbf{a}^\top\mathbf{X} + b$, with parameters $\mathbf{a}^\top \in \mathbb{R}^d$ and $b \in \mathbb{R}$, the mean and variance of the scalar output are then obtained as follows:

$$\mu_Y = \mathbb{E}[Y] = \mathbb{E}[\mathbf{a}^\top\mathbf{X} + b] = \mathbf{a}^\top\boldsymbol{\mu}_{\mathbf{X}} + b \equiv g(\boldsymbol{\mu}_{\mathbf{X}}), \qquad \sigma^2_Y = \mathrm{Var}(\mathbf{a}^\top\mathbf{X}) = \mathbf{a}^\top\boldsymbol{\Sigma}_{\mathbf{X}}\mathbf{a}. \tag{1.68}$$

The variance expression highlights that the output uncertainty depends not only on the variances of the components of $\mathbf{X}$, but also on their covariances through the covariance matrix $\boldsymbol{\Sigma}_{\mathbf{X}}$.

Finally, in the more general case in which the function output is itself a random vector

$\mathbf{Y} = \mathbf{g}(\mathbf{X}) = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{Y} \in \mathbb{R}^{d'}$, $\mathbf{A} \in \mathbb{R}^{d' \times d}$, and $\mathbf{b} \in \mathbb{R}^{d'}$, the mean vector and the covariance matrix of the function outputs are obtained as follows:

$$\boldsymbol{\mu}_{\mathbf{Y}} = \mathbb{E}[\mathbf{Y}] = \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b} \equiv \mathbf{g}(\boldsymbol{\mu}_{\mathbf{X}}), \qquad \boldsymbol{\Sigma}_{\mathbf{Y}} = \mathrm{Cov}(\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}_{\mathbf{X}}\mathbf{A}^{\top}. \tag{1.69}$$

1.5 Gaussian and Multivariate Gaussian Distributions

The *Gaussian distribution*, also known as the *normal distribution*, is one of the most important probability distributions in statistics and machine learning. It provides a simple yet powerful model for continuous random variables. It is present in many applications due to the central limit theorem[1]. Moreover, its mathematical properties make it particularly convenient for analytical calculations and numerical inference.

Univariate Gaussian.

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \tag{1.70}$$

which is denoted compactly as $X \sim \mathcal{N}(\mu, \sigma^2)$, where the $\sim$ symbol reads as *distributed like*.

The Gaussian distribution is completely characterized by its first two moments, as specifying μ and σ^2 uniquely determines the PDF. The mean μ determines the location of the distribution, which coincides with the mode where the distribution attains its maximum. The density is symmetric about μ, and the variance σ^2 controls its dispersion. Increasing σ^2 produces a broader distribution, whereas decreasing it concentrates the probability mass around the mean.

By introducing the following affine transformation:

$$Z = \frac{X - \mu}{\sigma}, \tag{1.71}$$

any Gaussian random variable can be transformed into a *standard* normal random variable $Z \sim \mathcal{N}(0, 1)$, defined as follows:

$$p(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right). \tag{1.72}$$

This transformation is commonly referred to as *standardization*. Figure 1.3 shows an exem-

[1] **Statement:** The distribution of sample means approaches a normal distribution as the sample size gets larger, regardless of the original distribution of the population.

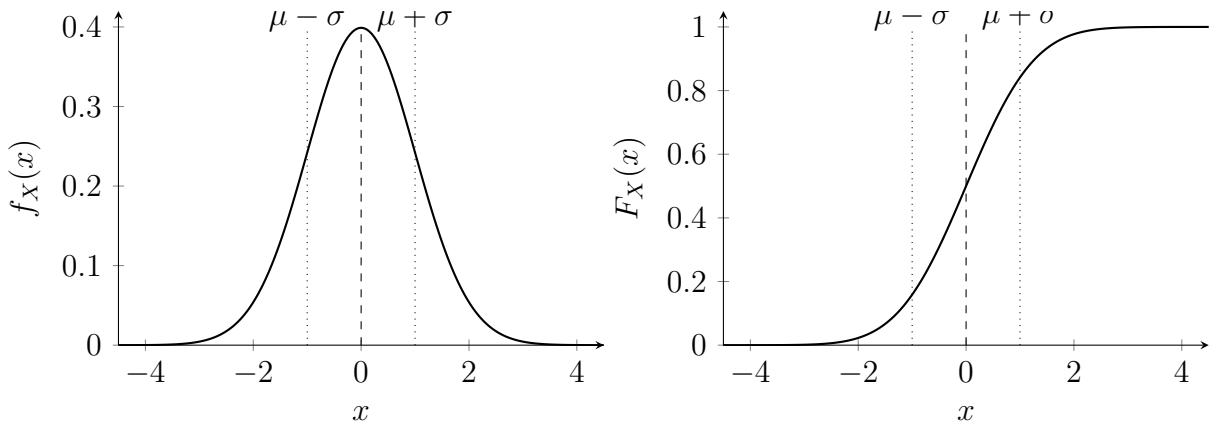


Figure 1.3: Probability density function and cumulative distribution function for a standard univariate Gaussian ($\mu = 0$ and $\sigma = 1$).

plary PDF and CDF for $\mu = 0$ and $\sigma = 1$. The inverse transformation can be obtained as $X = \mu + \sigma Z$.

Multivariate Gaussian.

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right], \tag{1.73}$$

which is denoted compactly as $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Here, $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{d\times d}$ are respectively the mean vector and covariance matrix.

A geometrical interpretation is provided by the factor $(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})$, which is known as squared *Mahalanobis distance*. Differently from the ordinary Euclidean distance, the Mahalanobis distance accounts for the covariance structure of the random variables. Directions characterized by large variance contribute less to the distance, whereas deviations along directions with small variance contribute more strongly to the distance.

Sets of points sharing the same probability density feature $(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = r^2$, where r is a positive constant. These sets correspond to d-dimensional ellipsoids centered at $\boldsymbol{\mu}$. The orientation and axis lengths of these ellipsoids are determined by the eigenvectors and eigenvalues of the covariance matrix. The eigenvectors define the principal directions of the Gaussian distribution, while the eigenvalues determine the variance along these directions, i.e., the length of the ellipsoids axes. Large eigenvalues correspond to directions along which the distribution is more dispersed.

1.5.1 Properties of the Gaussian Distribution

1. **Completely defined by $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.**

2. **Gaussian marginal distributions.** The marginal PDFs of a Gaussian distribution are also Gaussian:
$$X_i \sim \mathcal{N}(\mu_i, \Sigma_{ii}), \qquad i = 1, \ldots, d. \tag{1.74}$$

3. **Uncorrelated Gaussian variables are also statistically independent.** Moreover, statistically independent Gaussian variables allow factorizing joint Gaussian distributions as follows:
$$p(\mathbf{X}) = \prod_{j=1}^{d} p(x_j) = \prod_{j=1}^{d} \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left[-\frac{(x_j - \mu_j)^2}{2\sigma_j^2}\right]. \tag{1.75}$$

4. **Central limit theorem.** The distribution of the sample mean of independent and identically distributed random variables approaches a normal distribution as the sample size increases. This theorem highlights why aggregate quantities arising from the combination of many independent random contributions (as the outcomes of real-life phenomena) are often described as normally distributed, regardless of the distribution of the population.

5. **Linear Functions of Gaussian Variables are Gaussian.** Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_\mathbf{X}, \boldsymbol{\Sigma}_\mathbf{X})$ be a Gaussian random vector. For an affine transformation $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ with deterministic matrix $\mathbf{A} \in \mathbb{R}^{d' \times d}$ and deterministic vector $\mathbf{b} \in \mathbb{R}^{d'}$, the resulting random vector is also Gaussian with marginal distribution $\mathbf{Y} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu}_\mathbf{X} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}_\mathbf{X}\mathbf{A}^\top)$, and transformed mean vector $\boldsymbol{\mu}_\mathbf{Y} = \mathbb{E}[\mathbf{Y}] = \mathbf{A}\boldsymbol{\mu}_\mathbf{X} + \mathbf{b}$ and covariance matrix $\boldsymbol{\Sigma}_\mathbf{Y} = \mathrm{Cov}(\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}_\mathbf{X}\mathbf{A}^\top$. Furthermore, $\mathbf{X}$ and $\mathbf{Y}$ are jointly Gaussian according to:
$$\begin{bmatrix}\mathbf{X}\\ \mathbf{Y}\end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\boldsymbol{\mu}_\mathbf{X}\\ \mathbf{A}\boldsymbol{\mu}_\mathbf{X} + \mathbf{b}\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_\mathbf{X} & \boldsymbol{\Sigma}_\mathbf{X}\mathbf{A}^\top\\ \mathbf{A}\boldsymbol{\Sigma}_\mathbf{X} & \mathbf{A}\boldsymbol{\Sigma}_\mathbf{X}\mathbf{A}^\top\end{bmatrix}\right), \tag{1.76}$$
with cross-covariance matrix given by $\mathrm{Cov}(\mathbf{X}, \mathbf{Y}) = \mathrm{Cov}(\mathbf{X}, \mathbf{A}\mathbf{X} + \mathbf{b}) = \boldsymbol{\Sigma}_\mathbf{X}\mathbf{A}^\top$.

1.5.2 Measurement Noise

A typical application of the properties above is the modeling of noisy measurements. Measurements are generally affected by uncertainty, which can be modeled as the sum of the

ideal measurement and an additive noise term. A common assumption is that the *measurement noise* $\mathbf{v} \in \mathbb{R}^{n_y}$ is Gaussian and independent of the underlying quantity being measured $\mathbf{Y} \in \mathbb{R}^{n_y}$. In the multivariate case, we can express the noisy measurements as follows:

$$\mathbf{Y}_{\text{meas}} = \mathbf{Y} + \mathbf{v}, \qquad \mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{1.77}$$

where, for simplicity, the measurement noise is assumed to have zero mean, although a non-zero mean could be introduced to explicitly account for systematic bias, and covariance matrix $\mathbf{R} \in \mathbb{R}^{n_y \times n_y}$. If we assume that the noise-free measurement is described by a Gaussian random vector $\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{Y}}, \boldsymbol{\Sigma}_{\mathbf{Y}})$, or equivalently by an affine transformation of a Gaussian random vector, then $\mathbf{Y}_{\text{meas}}$ is the sum of independent Gaussian random vectors and is therefore also Gaussian, with the following distribution:

$$\mathbf{Y}_{\text{meas}} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{Y}}, \boldsymbol{\Sigma}_{\mathbf{Y}} + \mathbf{R}). \tag{1.78}$$

This shows that additive independent Gaussian measurement noise leaves the mean unchanged, while the covariance matrix of the noisy measurement is obtained by adding the noise covariance matrix $\mathbf{R}$ to the covariance matrix of the noise-free measurement. If the measurement errors are uncorrelated, the additional covariance matrix $\mathbf{R}$ is diagonal, with diagonal entries representing the variances of the individual measurement errors.

In many cases, the noise-free measurement is itself modeled as a linear transformation of another Gaussian random vector $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{X}}, \boldsymbol{\Sigma}_{\mathbf{X}})$. For instance, this is the case in Kalman filtering, where the random vector $\mathbf{X} \in \mathbb{R}^{n_x}$ represents the state of a system, from which measurements are extracted as according to the following observation model:

$$\mathbf{Y} = \mathbf{H}\mathbf{X}, \tag{1.79}$$

where $\mathbf{H} \in \mathbb{R}^{n_y \times n_x}$ is called observation matrix. Therefore, the noisy measurement can be written as $\mathbf{Y}_{\text{meas}} = \mathbf{H}\mathbf{X} + \mathbf{v}$. Since $\mathbf{H}\mathbf{X}$ is Gaussian by the properties of linear transformations of Gaussian random vectors, and $\mathbf{v}$ is an independent Gaussian random vector, $\mathbf{Y}_{\text{meas}}$ is also Gaussian, with the following distribution:

$$\mathbf{Y}_{\text{meas}} = \mathbf{H}\mathbf{X} + \mathbf{v} \sim \mathcal{N}(\mathbf{H}\boldsymbol{\mu}_{\mathbf{X}}, \mathbf{H}\boldsymbol{\Sigma}_{\mathbf{X}}\mathbf{H}^{\top} + \mathbf{R}). \tag{1.80}$$

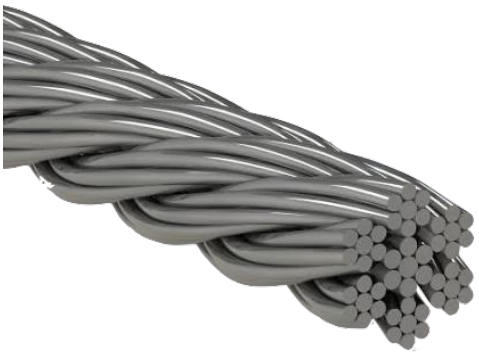


Figure 1.4: Steel cable made from dozens of individual wires.

1.5.3 Example: Steel Cable Resistance

Consider the following example adapted from [1]. Let $X_i \sim \mathcal{N}(10\,\text{kN}, 9\,\text{kN}^2)$, $i = 1, \ldots, N_{\text{wires}}$, represent the resistance of the individual steel wires composing a cable. Consider a cable made of $N_{\text{wires}} = 50$ wires, and assume that its total resistance is given as follows:

$$X_{\text{cable}} = \sum_{i=1}^{N_{\text{wires}}} X_i. \tag{1.81}$$

Since the sum of jointly Gaussian random variables is also Gaussian, the cable resistance can be written as $X_{\text{cable}} \sim \mathcal{N}(\mu_{\text{cable}}, \sigma^2_{\text{cable}})$. Its statistical properties depend on the covariance structure of the wire resistances, see Eq. (1.68), and in particular on the correlation coefficients $\rho_{i,j}$, $\forall i \neq j$ with $i, j = 1, \ldots, N_{\text{wires}}$.

The mean of the cable resistance is given as follows:

$$\mu_{\text{cable}} = \mathbb{E}\left[\sum_{i=1}^{N_{\text{wires}}} X_i\right] = \sum_{i=1}^{N_{\text{wires}}} \mathbb{E}[X_i] = N_{\text{wires}} \times 10\ \text{kN} = 500\ \text{kN},$$

which does not depend on the correlations among the individual wire resistances. The variance of the cable resistance, instead depends on the covariances between the individual wires, as follows:

$$\sigma^2_{\text{cable}} = \text{Var}\left(\sum_{i=1}^{N_{\text{wires}}} X_i\right) = \sum_{i=1}^{N_{\text{wires}}} \text{Var}(X_i) + 2\sum_{i<j} \text{Cov}(X_i, X_j).$$

Since all wires have standard deviation $\sigma_{X_i} = 3$ kN, the covariance between two wires is $\text{Cov}(X_i, X_j) = \rho_{i,j}\sigma^2_{X_i} = 9\rho_{i,j}$ kN^2. To illustrate, consider two limit cases:

1. **Independent wire resistances:** $\rho_{i,j} = 0$. All the non-diagonal terms of the covari-

ance matrix $\mathbf{\Sigma_X}$ are zero, and we obtain:

$$\sigma^2_{\text{cable}} = N_{\text{wires}} \times 9 \text{ kN}^2 = 450 \text{ kN}^2.$$

2. **Perfectly correlated wire resistances:** $\rho_{i,j} = 1$. All the non-diagonal terms of the covariance matrix $\mathbf{\Sigma_X}$ are 9 kN^2, and we obtain:

$$\sigma^2_{\text{cable}} = N^2_{\text{wires}} \times 9 \text{ kN}^2 = 22,500 \text{ kN}^2.$$

Intuitively, if the resistances of the individual wires are independent, positive and negative deviations from their respective means tend to compensate each other, yielding a relatively small uncertainty in the total cable resistance. On the other hand, when the wire resistances are perfectly correlated, deviations occur in the same direction across all wires and therefore uncertainties reinforce each other. In the independent case, the standard deviation of the total resistance increases with the square root of the number of wires; if wire resistances are perfectly correlated, it instead increases linearly with the number of wires.

1.6 Bayes

Bayes' theorem provides the basis for updating probabilistic beliefs when new information becomes available. In engineering applications, it provides a framework for incorporating prior knowledge about an uncertain quantity and updating this knowledge in light of observations.

Interpretation of Probability. Probability can be interpreted in different ways depending on the nature of the uncertainty being considered. In a *frequentist interpretation*, probability is associated with the relative frequency with which an event occurs over a sufficiently large number of repeated experiments. In a *Bayesian interpretation*, probability quantifies the degree of belief or uncertainty associated with a proposition given the available information.

Intuitively, the frequentist description is most suitable for aleatoric uncertainty, while the Bayesian description is most useful for dealing with epistemic uncertainty. In the Bayesian setting, probabilities are *updated* when additional information becomes available. The latter interpretation is the dominant perspective in engineering applications involving data assimilation, where uncertainty about unknown quantities of interest is reduced as new observations become available.

Bayes' Rule. The theorem reads as follows:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \tag{1.82}$$

which is a direct consequence of using the chain rule of probability in Eq. (1.20) to break the joint $P(A \cap B)$ into two different products of a conditional and its associated marginal probability. It relates two conditional probabilities that describe the same joint event from opposite directions. In particular, $P(B \mid A)$ describes the probability of observing B assuming that hypothesis A is true, whereas $P(A \mid B)$ describes the probability of A after observing B. Herein: $P(A \mid B)$ is the *posterior belief*, encoding the knowledge for A after observing B; $P(A)$ is the *prior belief*, encoding the knowledge for A before observing B. This prior knowledge may be based on heuristics, such as expert opinions, or derived from historical data and in situ inspections. In the cases where data are acquired *sequentially*, the posterior knowledge from the previous time becomes the prior for the current time; $P(B \mid A)$ is the *likelihood function*, quantifying how compatible the observation B is with the hypothesis A. It serves as the mechanism informing the posterior distribution about the available data. In particular, it works by describing the probability of observing event B given A, exploiting a suitable forward map linking A and B; finally, $P(B)$ is the *evidence* or *marginal likelihood*, measuring the probability of observing B independently of which hypothesis is true. Since this last term only acts as a normalization factor ensuring that the posterior probabilities sum to one to form a valid probability distribution, Bayes' theorem can be summarized conceptually as follows:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}. \tag{1.83}$$

Moreover, exact computation of the evidence term is usually very challenging, as it often involves multi-dimensional nonlinear integrals. For this reason, the posterior is usually approximated through numerical techniques.

Updating Mechanism. To better visualize how the updating mechanism operates, consider to infer among K competing hypotheses $A_1, \ldots, A_K$. For example, each hypothesis A_k, $k = 1, \ldots, K$, may represent a different material condition, while B may represent a set of measurements collected from the physical system. Each hypothesis A_k is initially assigned a prior probability $P(A_k)$. The observation of B then provides evidence in favor of some hypotheses and against others through the corresponding likelihoods $P(B \mid A_k)$. The posterior

probability combines these two sources of information, as follows:

$$P(A_k \mid B) = \frac{P(B \mid A_k)P(A_k)}{P(B)} = \frac{P(B \mid A_k)P(A_k)}{\sum_{j=1}^{K} P(B \mid A_j)P(A_j)}, \qquad k = 1, \ldots, K, \tag{1.84}$$

where normalization ensures that $\sum_{k=1}^{K} P(A_k \mid B) = 1$.

1.6.1 Example: A Diagnostic Test for a Rare Disease

The following example has been adapted from Blitzstein and Hwang [2]. Consider a rare disease, which occurs in only 1% of the population. Patient Tom undergoes a diagnostic test for this disease and receives a positive result.

Two possible associated events are as follows:

$$D = \{\text{Tom has the disease}\}, \qquad T = \{\text{the test result is positive}\}.$$

Assume that the diagnostic test is 95% accurate, meaning it correctly identifies 95% of diseased individuals and also correctly identifies 95% of healthy individuals. In probability notation, we have that $P(T \mid D) = 0.95$ and $P(T^c \mid D^c) = 0.95$, respectively. The first probability is the *sensitivity* of the test (or true positive detection rate), whereas the second is its *specificity* (or true negative detection rate).

Given Tom's positive result, what is the probability that he actually has the disease?

The required probability is $P(D \mid T)$. Bayes' rule gives:

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T)}.$$

The overall probability of obtaining a positive test $P(T)$ can be decomposed according to whether the person is diseased or healthy, as follows:

$$P(T) = P(T \mid D)P(D) + P(T \mid D^c)P(D^c).$$

Since the specificity is 0.95, the probability that a healthy person nevertheless obtains a positive result is $P(T \mid D^c) = 1 - 0.95 = 0.05$. Furthermore, experimental evidence about the disease occurring in 1% of the population informs the prior probabilities $P(D) = 0.01$

and $P(D^c) = 0.99$. By substituting:

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T \mid D)P(D) + P(T \mid D^c)P(D^c)} = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} = \frac{0.0095}{0.059} \approx 0.161.$$

A positive test raises (with respect to the prior) the probability that Tom has the disease, but the probability is still only about 16% even though the test is 95% accurate. The reason is that the disease is uncommon, so the low prior probability about the prevalence of the disease is playing a key role in modulating the posterior probability. This outcome is the result of two weighted contributions, associated with the rarity of the disease and the rarity of a mistaken test result.

Interpretation with a population of 10^4. The same result can be understood without directly applying Bayes' formula. Imagine testing 10^4 people. Since 1% of the population has the disease, we expect

$$\text{100 diseased individuals,}$$
$$\text{9900 healthy individuals.}$$

Let us focus on those individuals who test positive, i.e., conditioning on a positive test result. Because the sensitivity is 95%, approximately 95 of the 100 diseased individuals will test positive, while about 5 will test negative. For the 9900 healthy individuals, a specificity of 95% means that about 95% will correctly test negative. The remaining 5% will produce false positive results. Hence, $0.05 \times 9900 = 495$ healthy individuals are expected to test positive. The positive-test group therefore contains approximately $95 + 495 = 590$ people, but only 95 of them actually have the disease.

Once again, we obtain:

$$P(D \mid T) \approx \frac{95}{590} \approx 0.161$$

. Most people who test positive for the disease don't actually have the disease, highlighting the importance of the *base rate*. Even a test with high sensitivity and specificity can produce a relatively low probability of disease after a positive result when the disease itself is rare.

1.6.2 Bayes' Theorem for Random Variables.

The interpretation described above similarly applies to random variables, as follows:

$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)} = \frac{p(y \mid x)p(x)}{\int_{-\infty}^{\infty} p(y \mid x)p(x)\,\mathrm{d}x}. \tag{1.85}$$

In this version, the prior density $p(x)$ represents the uncertainty about X before observing $Y = y$, while the likelihood $p(y \mid x)$ describes how probable the observation is for different values of x. The posterior density $p(x \mid y)$ combines the two and represents the updated uncertainty after observing $Y = y$.

Engineering Interpretation. In engineering, the Bayes theorem is usually employed to update knowledge about uncertain parameters $\boldsymbol{\theta} \in \mathbb{R}^d$, modeled as random variables $\boldsymbol{\Theta} \sim p(\boldsymbol{\theta})$, after assimilating measured data $\mathbf{y} \in \mathbb{R}^{N_y}$ from an uncertain experiment, such that $\mathbf{Y} \sim p(\mathbf{y})$, as follows:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})} = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}}. \tag{1.86}$$

Here, $p(\boldsymbol{\theta})$ represents prior knowledge about the uncertain parameters, while $p(\mathbf{y} \mid \boldsymbol{\theta})$ describes the probability of obtaining the measured data for a given parameter value. The resulting posterior density $p(\boldsymbol{\theta} \mid \mathbf{y})$ quantifies the updated knowledge about the parameters after assimilating the measurements. The evidence term at the denominator ensures that the posterior density integrates to one and quantifies how well the probabilistic model predicts the observed data.

A Bayesian inverse problem typically involves three main ingredients: (i) a forward model $\mathcal{M} : \mathbb{R}^d \mapsto \mathbb{R}^{N_y}$, which maps a set of input parameters $\boldsymbol{\theta}$ into model predictions $\widehat{\mathbf{y}} = \mathcal{M}(\boldsymbol{\theta})$; (ii) the uncertain parameters $\boldsymbol{\theta}$ to be inferred; and (iii) experimental or observational data $\mathbf{y}$. The forward model $\mathcal{M}$ is a mathematical representation of the physical system under consideration and may be statistical, data-driven, mechanistic, or a combination of thereof. Often, this involves solving a differential problem through an analytical or numerical simulator. Because such models can be computationally expensive, data-driven surrogate models are often employed to approximate the forward map and reduce the computational cost of Bayesian inference.

The model output $\widehat{\mathbf{y}} = \mathcal{M}(\boldsymbol{\theta})$ is an approximation of the real-world observation $\mathbf{y}$, which

is interpreted as follows:

$$\mathbf{Y} = \mathcal{M}(\boldsymbol{\Theta}) + \mathbf{Y}_{\varepsilon}, \tag{1.87}$$

where $\mathbf{Y}_{\varepsilon} \in \mathbb{R}^d$ is an additive discrepancy term accounting for measurement errors and model inaccuracies. This discrepancy is modeled as a random variable $\mathbf{Y}_{\varepsilon} \sim p(\mathbf{y}_{\varepsilon})$, and is assumed to be independent of the model parameters $\boldsymbol{\Theta}$. For simplicity, it is also commonly assumed to have zero mean, although a non-zero mean could be introduced to explicitly account for systematic bias.

The goal of Bayesian model inversion is to infer the posterior distribution of $\boldsymbol{\theta}$ that quantifies the plausibility of different parameter values after observing the experimental data $\mathbf{y}$. Thanks to the assumption of independence between $\boldsymbol{\Theta}$ and $\mathbf{Y}_{\varepsilon}$, the residual $\mathbf{Y} - \mathcal{M}(\boldsymbol{\Theta})$ has the same distribution as the observation error $\mathbf{Y}_{\varepsilon}$. Therefore, the specification of the likelihood function depends on the probabilistic assumptions made regarding $\mathbf{Y}_{\varepsilon}$. A simple and widely used probabilistic model that can be used to describe experimental uncertainties is the *Gaussian* model $\mathbf{Y}_{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon})$, with zero mean and covariance matrix $\boldsymbol{\Sigma}_{\varepsilon} \in \mathbb{R}^{N_y \times N_y}$. Under this assumption, the likelihood function is also Gaussian and reads:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{(2\pi)^{N_y/2}|\boldsymbol{\Sigma}_{\varepsilon}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mathcal{M}(\boldsymbol{\theta}))^{\top}\boldsymbol{\Sigma}_{\varepsilon}^{-1}(\mathbf{y} - \mathcal{M}(\boldsymbol{\theta}))\right). \tag{1.88}$$

The quadratic term in the exponential is the squared Mahalanobis distance between the observed data and the model prediction, weighted by the covariance structure of the observational uncertainty. Deviations along directions associated with large uncertainty contribute less to the likelihood penalty, whereas deviations along directions characterized by small uncertainty contribute more strongly.

Combining this likelihood with the prior according to Bayes' theorem gives the posterior distribution $p(\boldsymbol{\theta} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})$. Point estimates, such as the maximum a-posteriori estimate or the posterior mean, can subsequently be computed from this distribution.

1.6.3 Bayesian Gaussian Inference

The properties of Gaussian random variables introduced in the previous sections lead to an important result for Bayesian inference.

Remark. The conditional distribution of jointly Gaussian random variables is Gaussian.

This property provides a simple setting in which the effect of an observation on our knowledge of an uncertain quantity can be derived analytically.

Consider a jointly Gaussian random vector $\mathbf{X} \in \mathbb{R}^d$ partitioned into two subsets,

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_i \\ \mathbf{X}_j \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_i \\ \boldsymbol{\mu}_j \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_i & \boldsymbol{\Sigma}_{ij} \\ \boldsymbol{\Sigma}_{ji} & \boldsymbol{\Sigma}_j \end{bmatrix} \right), \tag{1.89}$$

where $\boldsymbol{\Sigma}_{ij}$ describes the covariance between the two subsets and $\boldsymbol{\Sigma}_{ji} = \boldsymbol{\Sigma}_{ij}^\top$.

Following the definition of conditional probability, the conditional density of $\mathbf{X}_i$ given an observation $\mathbf{X}_j = \mathbf{x}_j$ is

$$p_{\mathbf{X}_i|\mathbf{x}_j}(\mathbf{x}_i|\mathbf{x}_j) = \frac{p_{\mathbf{X}_i,\mathbf{X}_j}(\mathbf{x}_i, \mathbf{x}_j)}{p_{\mathbf{X}_j}(\mathbf{x}_j)}. \tag{1.90}$$

For jointly Gaussian random variables, this conditional distribution is also Gaussian $\mathbf{X}_i|\mathbf{x}_j \sim \mathcal{N}\left(\boldsymbol{\mu}_{i|j}, \boldsymbol{\Sigma}_{i|j}\right)$, where the conditional mean and covariance are given as follows:

$$\boldsymbol{\mu}_{i|j} = \boldsymbol{\mu}_i + \boldsymbol{\Sigma}_{ij}\boldsymbol{\Sigma}_j^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_j), \tag{1.91}$$

$$\boldsymbol{\Sigma}_{i|j} = \boldsymbol{\Sigma}_i - \boldsymbol{\Sigma}_{ij}\boldsymbol{\Sigma}_j^{-1}\boldsymbol{\Sigma}_{ji}. \tag{1.92}$$

The conditional Gaussian distribution can be interpreted directly in Bayesian terms. Suppose that $\mathbf{X}_i$ represents an uncertain quantity of engineering interest and that $\mathbf{X}_j = \mathbf{x}_j$ represents an observation that provides information about it. The marginal distribution $p_{\mathbf{X}_i}(\mathbf{X}_i)$ represents the prior knowledge about $\mathbf{X}_i$, whereas the conditional distribution $p_{\mathbf{X}_i|\mathbf{x}_j}(\mathbf{x}_i|\mathbf{x}_j)$ represents the updated knowledge after observing $\mathbf{X}_j = \mathbf{x}_j$, i.e., the posterior distribution.

The observation changes both the expected value and the uncertainty associated with the quantity of interest. The conditional mean starts from the prior mean $\boldsymbol{\mu}_i$ and is shifted according to the discrepancy between the observation $\mathbf{x}_j$ and its prior mean $\boldsymbol{\mu}_j$. The magnitude and direction of this correction are determined by the covariance $\boldsymbol{\Sigma}_{ij}$ between the quantity of interest and the observation. On the other hand, the conditional covariance is smaller than or equal to the prior covariance, reflecting the reduction in uncertainty resulting from the additional information provided by the observation.

The Gaussian case therefore provides an analytical (closed-form) example of the general Bayesian updating principle introduced in the previous section. This result is particularly important, as many estimation and data assimilation methods rely on Gaussian assumptions. Indeed, the analytical tractability of Gaussian conditioning provides the theoretical foundation for methods such as Gaussian process regression and Kalman filtering.

Scalar Example. The result becomes particularly intuitive when considering two jointly Gaussian scalar random variables X_1 and X_2:

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \right), \tag{1.93}$$

where ρ is the correlation coefficient between X_1 and X_2. Conditioning on the observation $X_2 = x_2$ gives $X_1|x_2 \sim \mathcal{N}\left(\mu_{1|2}, \sigma_{1|2}^2\right)$.

The statistics $\mu_{1|2}$ and $\sigma_{1|2}^2$ can be directly computed by applying Eq. (1.91) and Eq. (1.92), respectively, as follows:

$$\mu_{1|2} = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2), \tag{1.94}$$

$$\sigma_{1|2}^2 = \sigma_1^2 - \frac{\rho\sigma_1\sigma_2 \cdot \rho\sigma_1\sigma_2}{\sigma_2^2} = \sigma_1^2(1 - \rho^2). \tag{1.95}$$

Note how x_2 modifies the expected value of X_1. If x_2 is larger than its prior mean and $\rho > 0$, the conditional mean of X_1 increases. Conversely, if $\rho < 0$, the conditional mean decreases. The strength of this correction is controlled by the ratio between the two variances. Moreover, uncertainty reduction in X_1 is proportional to $(1-\rho^2)$. The stronger the correlation among X_1 and X_2, the greater the reduction in uncertainty. In the limit case of $\rho = 0$, then $\mu_{1|2} = \mu_1$ and $\sigma_{1|2}^2 = \sigma_1^2$, i.e., zero correlation implies statistical independence for jointly Gaussian variables. At the opposite extreme, if $|\rho| = 1$, the conditional variance becomes zero, meaning that X_1 can be determined exactly from the observation of X_2.

Bibliography

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